

Introduction to Option Pricing

Table of contents

1	Start with the right to choose	2
2	Payoff is not profit	3
2.1	Example 1: a call with a \$100 strike	3
2.2	Example 2: a put protects against a lower bond price	3
2.3	The words “in the money” do not mean “profitable”	4
3	From possible payoffs to a price today	5
4	A one-period binomial tree	5
4.1	Example 3: calculate the premium, one step at a time	6
4.2	Why 60% is not a forecast	7
5	Build the same payoff without buying the option	7
6	Backward induction	8
6.1	Example 4: roll a two-period tree backward	9
6.2	If early exercise is allowed	9
7	Why this matters for fixed income	10
7.1	Example 5: who benefits from the embedded option?	11
8	Common points of confusion	11
9	Check your understanding	12

This appendix assumes no previous knowledge of options. You only need the idea that a dollar received later is worth less today when interest rates are positive. Read the payoff examples first, then follow the pricing calculation. The replication section explains *why* the pricing rule works; no calculus is required.

By the end, you should be able to distinguish a call from a put, separate payoff from profit, price a simple option in a two-state model, and explain why an issuer's call right reduces a bondholder's value.

1 Start with the right to choose

Imagine paying for the right to buy a bond for \$100 one year from now. If the bond is then worth \$110, buying it for \$100 is attractive. If it is worth \$90, you can let the right expire and buy the cheaper bond in the market instead. You have a choice; a forward contract would commit you to the agreed trade.

The choice is valuable, so the seller normally charges an upfront **premium**. That premium is the option's price. The buyer can decline to exercise, but does not get the premium back. The seller, also called the **writer**, must honor valid exercise under the contract.

Term	Plain-language meaning
Underlying	The asset or reference on which the option is based; here, a bond.
Call	The holder's right to buy the underlying at the agreed price.
Put	The holder's right to sell the underlying at the agreed price.
Strike, K	The price paid or received when the holder exercises.
Expiration, T	The last time at which the right can be exercised.
Premium	The option price paid by the buyer to the seller.
Exercise	Use the contractual right to buy or sell.

A **European** option can be exercised only at expiration. An **American** option can be exercised earlier as well. These names describe exercise timing, not where the option trades. Many embedded bond options allow exercise only on specified dates. We begin with European options to keep the decision simple.

All numbers below are illustrative. Prices are dollars for one unit of the bond; one option covers that unit. Ignore fees, bid-ask spreads, taxes, and default. In the pricing trees, the underlying bond makes no coupon or principal payment before the option expires, and borrowing and lending use the same risk-free rate. These assumptions are essential to the simple formulas.

2 Payoff is not profit

2.1 Example 1: a call with a \$100 strike

Suppose the call's supplied premium is \$3. We are studying its outcome here, not yet deciding whether \$3 is a fair price.

Bond price at expiration	Exercise the call?	Call payoff	Buyer's cash profit
\$90	No: buying for \$100 would overpay	\$0	−\$3
\$100	No economic gain from exercise	\$0	−\$3
\$102	Yes: buy for \$100, receive value \$102	\$2	−\$1
\$110	Yes: buy for \$100, receive value \$110	\$10	\$7

The **payoff** is the value obtained at expiration. **Cash profit** here means payoff minus the \$3 upfront premium, without adjusting for financing or the time value of money. Exercise can be worthwhile even when the overall trade loses money: at \$102 the \$2 payoff recovers part of the premium already spent.

With B_T denoting the bond price at expiration,

$$\text{Call payoff} = \max(B_T - K, 0), \quad \text{Cash profit} = \max(B_T - K, 0) - \text{premium}.$$

“Max” simply means choose the larger number. The holder can choose not to exercise, which prevents a negative payoff. The premium still creates a loss. Ignoring financing, the call buyer breaks even at $100 + 3 = 103$. If the premium is financed for one year at 2%, the terminal cost is $3(1.02) = 3.06$, so the financed break-even is 103.06 instead.

2.2 Example 2: a put protects against a lower bond price

Now consider a **separate** put with strike \$100 and supplied premium \$3. The put holder can sell for \$100 even if the market price is lower.

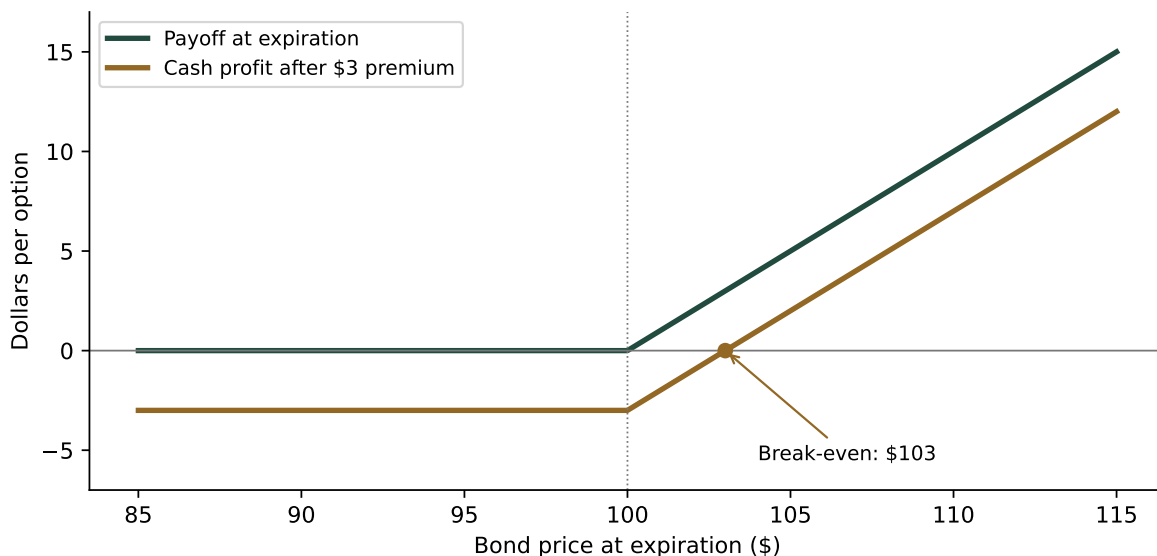


Figure 1: A call’s payoff cannot be negative, but the buyer can lose the premium. Cash profit here excludes financing.

$$\text{Put payoff} = \max(K - B_T, 0).$$

Bond price at expiration	Put payoff	Put buyer’s cash profit
\$90	$100 - 90 = \$10$	$10 - 3 = \$7$
\$100	\$0	−\$3
\$110	\$0	−\$3

A call benefits from a higher underlying price; a put benefits from a lower one. For a conventional fixed-rate bond, higher yields usually mean a lower bond price. A **put on the bond price** can therefore protect against a yield increase. Always identify the underlying: a call on an interest rate is a different contract from a call on a bond price.

2.3 The words “in the money” do not mean “profitable”

A call is **in the money** when the bond price exceeds its strike. A put is in the money when the bond price is below its strike. At the strike, either is **at the money**; on the other side it is **out of the money**. These terms compare the underlying price with the strike and ignore the premium.

Before expiration, an out-of-the-money option can still have value because the price may move before the exercise date. Its eventual payoff can be zero even though its current price is positive. Before-expiration option value and expiration payoff are different quantities.

3 From possible payoffs to a price today

We now ask a different question: **what premium is consistent with today's bond price and the possible future payoffs?** A simple average is not enough. We need both the time value of money and a pricing rule that respects the underlying bond's current price.

A **binomial tree** represents two possible next states: “up” and “down.” It is a teaching approximation, not a claim that markets have only two outcomes. **No arbitrage** means identical future cash flows must have the same price today; otherwise someone could buy the cheaper version and sell the dearer one.

4 A one-period binomial tree

The subscript 0 means today; u and d label the next up and down states. Suppose the bond is worth B_0 today. Over one period, it can move to:

- an up state, B_u ;
- a down state, B_d .

An option on the bond has a payoff in each state. For example, a call option with strike price K has payoff:

$$C_u = \max(B_u - K, 0), \quad C_d = \max(B_d - K, 0).$$

The option value today is the discounted expected payoff under **risk-neutral probabilities**:

$$C_0 = \frac{qC_u + (1 - q)C_d}{1 + r},$$

where q is the risk-neutral probability of the up state and r is the one-period risk-free rate. For a tradable bond with no payment during this interval,

$$q = \frac{(1 + r)B_0 - B_d}{B_u - B_d}, \quad B_d < (1 + r)B_0 < B_u.$$

The inequality places the risk-free growth of today's bond price between the two possible future prices. It gives $0 < q < 1$. If it fails, these inputs do not form a valid no-arbitrage two-state model. Do not fix the problem by clipping q to zero or one; reconsider the inputs.

The rate r is for **one tree interval**, not necessarily an annual quote. Dividing by $1 + r$ brings the next-period cash flow back to today. Any coupon within that interval must be included in the replication cash flows.

Intuition

The risk-neutral probability is not a forecast of what will actually happen. It is the probability that makes the model consistent with no-arbitrage pricing. We use it to price cash flows consistently with today's market prices.

4.1 Example 3: calculate the premium, one step at a time

Use a **new pricing example**: the bond costs \$100 today, is worth either \$110 or \$90 in one year, and pays nothing during that year. The call strike is \$100. The effective one-year risk-free rate is 2%. The option premium is now unknown.

Step 1 — Calculate the payoffs. The call pays \$10 in the up state and \$0 in the down state.

Step 2 — Find the pricing weight.

$$q = \frac{1.02(100) - 90}{110 - 90} = \frac{12}{20} = 0.60.$$

Check that the same weight prices the bond itself:

$$\frac{0.60(110) + 0.40(90)}{1.02} = \frac{102}{1.02} = 100.$$

Step 3 — Weight the option payoffs and discount.

$$C_0 = \frac{0.60(10) + 0.40(0)}{1.02} = \frac{6}{1.02} \approx 5.8824.$$

Thus the model-consistent premium is about **\$5.88 per option**. The weighted future payoff is \$6, but the price today is lower because of discounting. The earlier \$3 premium was an independent payoff illustration, not a price for this tree.

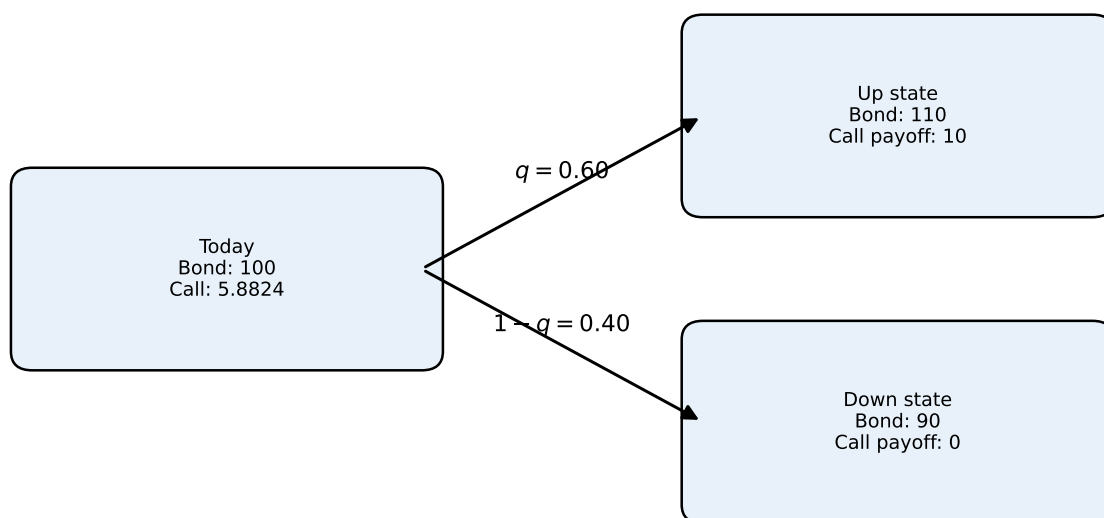


Figure 2: A one-period binomial tree for an option on a fixed-income underlying asset

4.2 Why 60% is not a forecast

The 60% weight makes the **current price** consistent with the model's future states. It does not assert that the bond has a 60% real-world chance of rising. A forecaster might assign another probability. Simply inserting that forecast probability and discounting at the risk-free rate would generally fail to price the underlying bond at \$100.

“Risk-neutral” does not mean investors ignore risk. It is a way of expressing prices after using the traded bond and risk-free borrowing to account for risk. The following replication calculation makes that concrete.

5 Build the same payoff without buying the option

A **replicating portfolio** is a combination of other traded positions with the same payoff as the option in every modeled state. Here we can buy part of a bond and borrow cash.

The call's payoff changes by \$10 between states while the bond changes by \$20. Buy half a bond:

$$\Delta = \frac{C_u - C_d}{B_u - B_d} = \frac{10 - 0}{110 - 90} = 0.5.$$

Half a bond is worth \$55 in the up state and \$45 in the down state. Borrow $45/1.02 = 44.1176$ today, so repayment next year is exactly 45.

Position at expiration	Up state	Down state
Half a bond	\$55	\$45
Repay the borrowing	−\$45	−\$45
Net payoff	\$10	\$0
Call payoff	\$10	\$0

The two payoffs match, so their prices must match:

$$C_0 = 0.5(100) - \frac{45}{1.02} = 50 - 44.1176 = 5.8824.$$

This is the same answer as risk-neutral pricing. If the option sold for \$7, for example, a trader in this frictionless model could sell it, buy the replicating portfolio for 5.8824, and keep the difference while the future cash flows cancel. This argument assumes short selling, borrowing, and trading are possible at the stated prices.

The hedge ratio Δ is 0.5 **for this node and this model**. In a multi-step tree it generally changes as the underlying moves; replication then requires rebalancing.

6 Backward induction

For options with several possible exercise dates, the tree has many steps. The pricing method works **backward**:

1. Build possible future states of the underlying asset.
2. Compute the option payoff at the final nodes.
3. Move one step backward by discounting the expected future option value.
4. If the option can be exercised early, compare the continuation value with the exercise value at each node.
5. Continue backward until reaching today's value.

This is called **backward induction**. It is central for valuing callable and putable bonds because the exercise decision must be checked at each possible future call or put date.

6.1 Example 4: roll a two-period tree backward

Let the bond start at \$100 and move up by 10% or down by 10% in **each** period. The risk-free rate is 2% per period and the European call strike is \$100. The bond pays no cash before option expiration. At the end of two periods:

Path	Bond price	Call payoff
Up, then up	$100(1.10)^2 = 121$	\$21
Up, then down	$100(1.10)(0.90) = 99$	\$0
Down, then up	$100(0.90)(1.10) = 99$	\$0
Down, then down	$100(0.90)^2 = 81$	\$0

The middle paths arrive at the same price, so the tree **recombines**. Because the proportional moves and interest rate are the same at each node, $q = (1.02 - 0.90)/(1.10 - 0.90) = 0.60$ at every step.

First calculate option values **one period before expiration**:

$$C_u = \frac{0.60(21) + 0.40(0)}{1.02} = 12.3529, \quad C_d = \frac{0.60(0) + 0.40(0)}{1.02} = 0.$$

Then move back to today:

$$C_0 = \frac{0.60(12.352941 \dots) + 0.40(0)}{1.02} = 7.2664.$$

Equivalently, the only positive terminal payoff has pricing weight $0.60^2 = 0.36$, so $C_0 = 0.36(21)/(1.02)^2$. Discount twice because two periods remain. Keep unrounded numbers during calculation.

6.2 If early exercise is allowed

For an American call, compare immediate exercise value with the value of continuing to hold the option. The holder chooses the **larger**:

$$\text{Option value} = \max(\text{exercise value}, \text{continuation value}).$$

At the up node in this example, immediate call exercise gives $110 - 100 = 10$, while continuation is 12.3529. The holder waits. This calculation concerns the option holder's choice. The minimum rule for a **callable bond investor** below reflects a different party making the decision: the issuer.

7 Why this matters for fixed income

For a plain fixed-rate bond, the cash flows are fixed, so we can usually value the bond directly from today's discount curve:

$$P = \sum_i CF_i P(0, t_i).$$

For a bond with an embedded option, the future cash flows depend on future rates. The tree must therefore generate future interest-rate states, future discount factors, and future bond values. At each exercise date, the model asks whether the option should be exercised.

For a callable bond, the issuer owns the option. At a future call date, the model produces a future yield curve $P(t, T)$. That curve is used to value the remaining coupon and principal payments:

$$V_{\text{continue}}(t) = \sum_i CF_i \times P(t, t_i).$$

The issuer compares this continuation value with the call price:

- the value of leaving the bond outstanding;
- the call price.

If rates have fallen, the old high-coupon bond is valuable to investors but expensive for the issuer to keep outstanding. The continuation value is likely to exceed the call price, so the issuer calls the bond. If rates have risen, the continuation value is usually below the call price, so the issuer does not call.

From the bondholder's perspective:

$$V(t) = \min(V_{\text{continue}}(t), \text{call price}).$$

The minimum appears because the issuer controls the call option. The bondholder receives the lower of keeping the valuable bond alive or being paid the call price.

Step	Valuation or decision
1. Generate the future curve	Obtain discount factors $P(t, T)$ for remaining payments.
2. Value remaining cash flows	Sum each promised payment times its discount factor.
3. Compare ex-coupon values	Compare continuation value with the redemption price.

Step	Valuation or decision
4. Continuation exceeds call price	Issuer calls; investor receives the call price.
5. Continuation is below call price	Issuer leaves the bond outstanding.
6. Values are equal	Issuer is indifferent in the frictionless model.

7.1 Example 5: who benefits from the embedded option?

Suppose a callable bond can be redeemed for \$101 immediately **after** a coupon has been paid. Compare the value of its remaining payments with that ex-coupon redemption price:

Future rate state	Value if bond continues	Issuer's choice	Investor's ex-coupon value
Lower rates	\$108	Call for \$101	\$101
Higher rates	\$96	Leave outstanding	\$96

The investor loses \$7 of continuation value in the lower-rate state. That \$7 is a **future state-specific reduction**, not today's embedded call value. Today's option cost requires modeling and discounting all relevant states. The current coupon is already paid and is not included a second time.

For a **puttable** bond, the investor owns the exercise right. With an ex-coupon put price of \$101, the investor can choose \$101 instead of continuing at \$96, but would keep the bond if continuation were \$108. Thus the puttable investor's node value is $\max(V_{\text{continue}}, 101)$.

This difference in control explains the price relationships for otherwise identical bonds:

$$V_{\text{callable}} = V_{\text{straight}} - V_{\text{issuer call}},$$

$$V_{\text{puttable}} = V_{\text{straight}} + V_{\text{investor put}}.$$

8 Common points of confusion

- **Premium versus strike:** the premium buys the right today; the strike is the contract price used upon exercise. They are different payments.
- **Payoff versus profit:** a positive payoff does not imply a profitable trade. Subtract the premium and, when relevant, financing and costs.
- **Price today versus payoff later:** discount future payoffs before comparing them with an upfront premium.

- **Probability versus pricing weight:** risk-neutral weights reproduce traded prices under the model; they are not forecasts of actual frequencies.
- **Bond price versus yield:** for conventional fixed-rate bonds they generally move in opposite directions. Name the option's underlying before interpreting it.
- **Maximum versus minimum:** an option holder chooses the better exercise outcome; a callable bond investor is exposed to the issuer choosing to call.

9 Check your understanding

1. A call has strike \$100 and costs \$3. The bond finishes at \$102. Does the holder exercise? What are payoff and cash profit before financing?
2. A put has strike \$100 and costs \$3. The bond finishes at \$94. What are its payoff, cash profit, and break-even bond price before financing?
3. In Example 3, why is the call worth \$5.8824 rather than \$6? Does $q = 0.60$ describe a 60% real-world chance of the up state?
4. In Example 5, who chooses whether to call, and why does the investor receive \$101 rather than \$108 in the lower-rate state?

i Worked answers

1. Yes. Exercise recovers \$2; cash profit is $2 - 3 = -1$. The premium is already spent, so a loss on the overall trade is no reason to forgo a positive payoff.
2. The put pays $100 - 94 = 6$; cash profit is $6 - 3 = 3$. Break-even is $100 - 3 = 97$.
3. The \$6 weighted payoff arrives in one year. Discounting at 2% gives $6/1.02 = 5.8824$. The 60% weight prices the underlying correctly; it is not a real-world forecast.
4. The issuer owns the right. Redeeming for \$101 is cheaper than keeping an obligation worth \$108 outstanding, so the investor's ex-coupon value is capped.

This is why the short-rate models discussed in [Lecture 7](#) are useful. They provide the future rate tree needed to value fixed-income options and bonds with embedded options.

i Bridge to Lecture 8

Continue with [Lecture 8](#) and its [callable-bond demo](#) to apply this logic to bonds with embedded options. The key idea will be the same: build future interest-rate states, value the bond at future nodes, apply the exercise rule, and work backward to today.